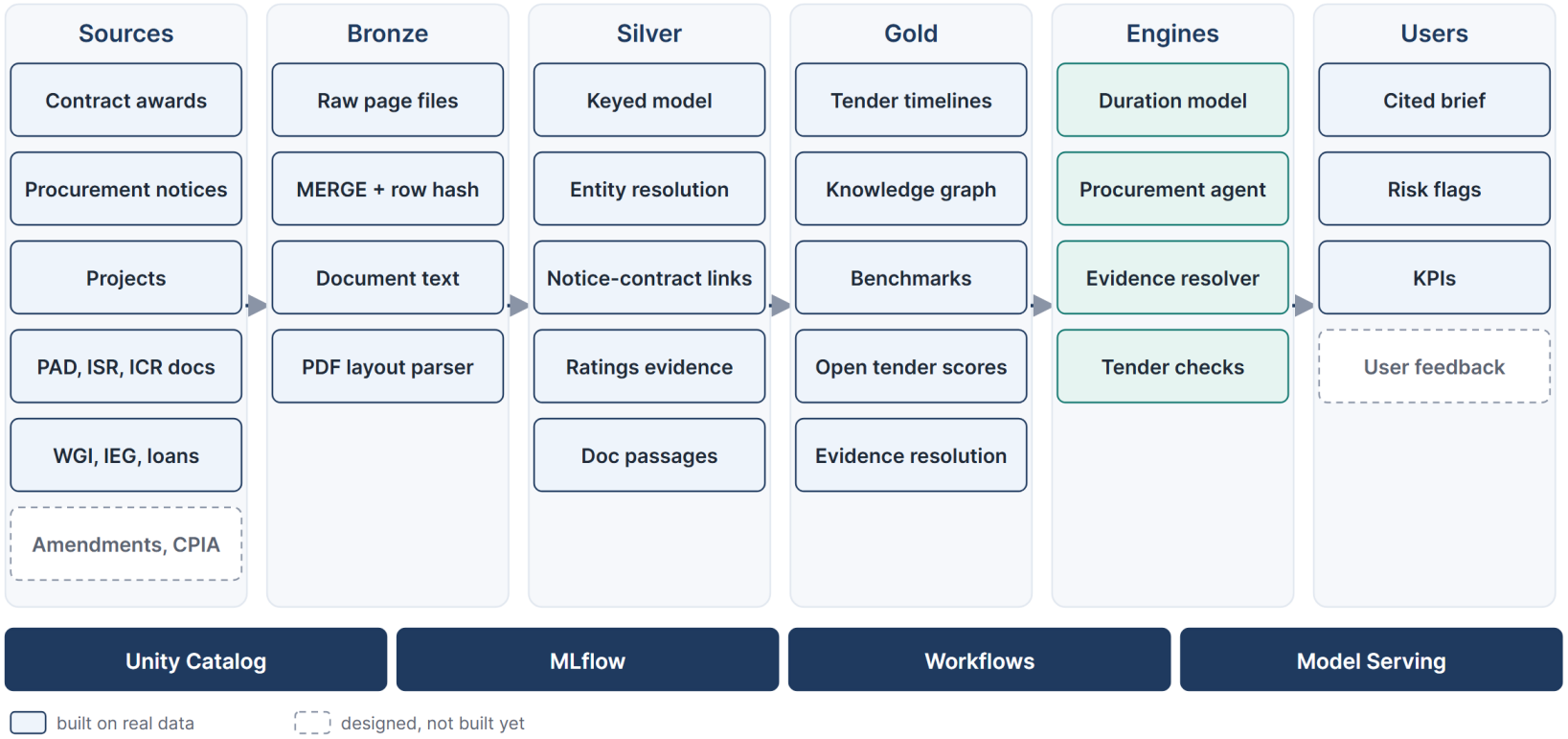
Procurement intelligence on real World Bank data

The registry says what happened, notices say how, completion reports say why. I joined them and built the platform.



282,890

contracts, linked to 379,820 notices and 3,768 projects

261,169

supplier firms resolved from 344,273 ids; precision 1.00, recall 0.90

75 vs 88 days

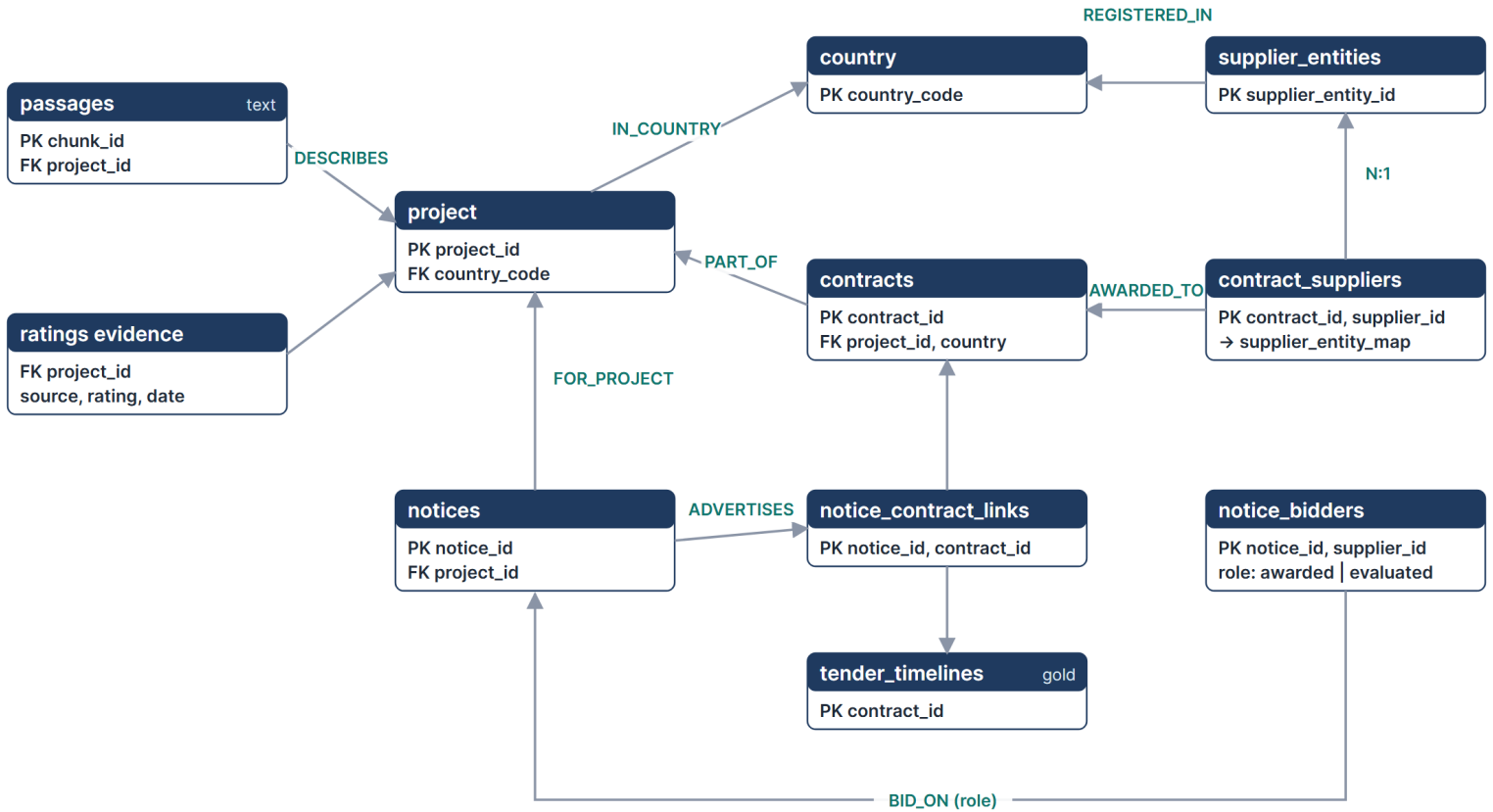
duration error on 6,624 test tenders: model trained before them vs the peer lookup

300 tenders

replayed as of their notice date to test the whole platform, with nothing later visible

One keyed data model, and a knowledge graph built from it

Grain and keys first; every graph edge comes from a key that was checked for duplicates and orphans



Entity resolution

- blocked by country; TF-IDF trigrams
- rules stop chaining; common names not merged
- P 1.00, R 0.90 on 97 pairs; audit 34 of 40

Tender timelines

- notice-contract links in 4 tiers, each audited
- 44.9% of open contracts timed; median 158 days
- joint-venture amounts kept once per contract

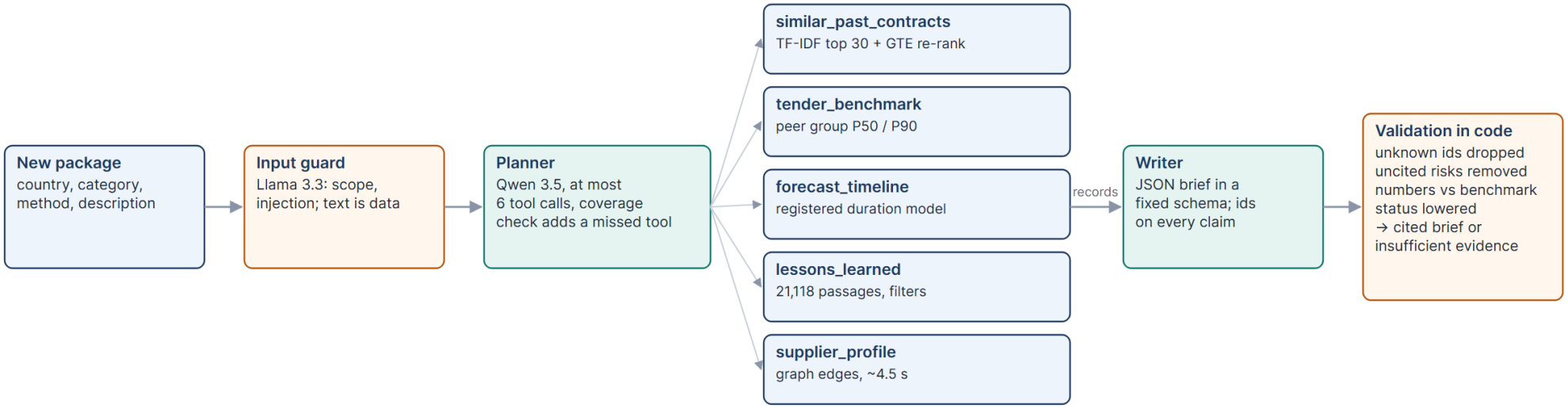
Knowledge graph in Delta

- 1.03M nodes, 2.28M edges, 0 dangling
- same answer from SQL joins and GraphFrames

Brief's example, contract row to ICR lesson: contract -PART_OF-> project -> completion-report passage of that project, or the nearest passages by embedding within country and sector. Dates decide whether a lesson explains the contract or is a precedent for a new one.

A calibrated duration model and an agent that must cite

Statistics come from models and tables; the LLM plans, retrieves and writes, and code checks what it wrote



Orange: deterministic or independent checks. Teal: LLM steps. One MLflow trace per run records the guard, every tool call, the records returned and the brief.

Duration model

- trained only on fully observable cohorts
- conformal P10-P90: 80.7% coverage
- package fields only: 91 vs 85 days for the peer group; with notice facts 60 (20 indicative cases)

Tender and execution checks

- 672 of 1,939 international tenders advertised < 30 business days: 4.9 vs 5.6 bidders
- 1,340 active projects: 73 red, 238 amber, reasons cited

Brief excerpt: rural roads, Kenya

Timeline: median 119 days to signing, P90 540.
[benchmark:country_method_category-5fa286a9]

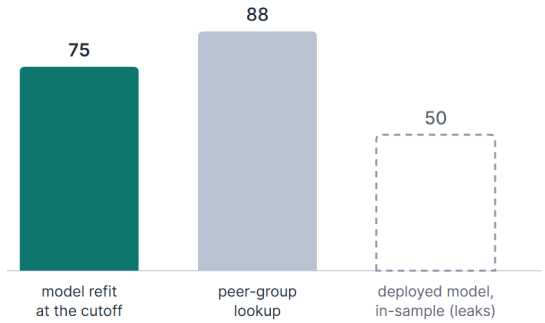
Risk: works awarded in the rainy season stall; sequence awards to the dry season.
[lesson_chunk:P123828-34118211-0110]

Trust: a backtest as of the notice date, flagged conflicts, stated gaps

300 real tenders replayed with everything later hidden; the deployed model's in-sample score is shown as a leak

Tender duration error, 6,624 test tenders

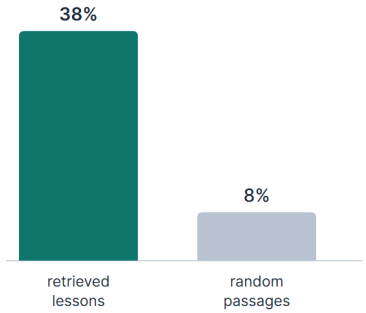
mean absolute error of P50, days; lower is better



The deployed @champion was refit on these tenders, so its 50 days is not a result.

Lessons that named a real problem

100 projects; did the project's own ICR report it? (LLM judge)



Yes or partly: 84% vs 22%. Judge agrees with an AI agent's blind spot-check on 13 of 15.

Backtest on 300 replayed tenders

- timeline error 87 days (95% CI 77-98) vs 100 for the peer lookup; slow-flag AUC 0.74 vs 0.58
- precedents: 77% of those shown relevant; 73% of packages get one, 19% none
- 30 full briefs: actual inside the cited range 20/30; unsupported claim 5/30
- labels: code checks, then 3 LLM judges, then an AI adjudicator with evidence

Guardrails: input guard 3/3; code drops ids no tool returned; 19 of 20 briefs judged grounded

35 of 68

projects IEG rated unsatisfactory still had a satisfactory last ISR (400 closed projects)

Conflict policy

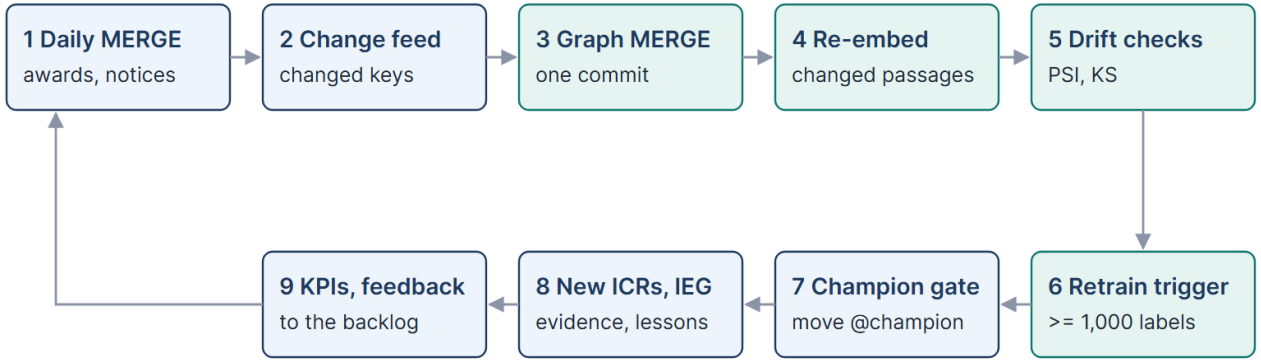
- outcome: IEG > ICR > last ISR > earlier ISR
- current status: newest ISR, marked stale when old
- both sides shown with a flag and confidence

Data gaps, handled openly

- amendments and milestones not public (STEP)
- plain chunking lost 26.8% of table rows; fixed
- awards from FY2020 only; 300 of ~8,100 ICRs

Rollout: a closed learning loop and a gated pilot

Incremental updates proven on real data; each pilot phase has a go/no-go threshold



Proven on a replay of real notices

- 934 inserts + 100 corrections -> 3,290 graph inserts, 50 deletes
- incremental result = full rebuild = production: 0 differences

KPIs, stored in pilot_kpis with baseline and target

- planning cycle time: 160 days today; -10% in pilot hubs vs similar hubs
- precision@5 >= 0.70; valid citations >= 0.95; freshness >= 0.95
- adoption >= 50% weekly; positive feedback >= 70%

0 Shadow, 4-6 weeks

answers logged, not shown; trust KPIs must pass

1 Assisted pilot, 3 months

2-3 regional hubs, 15-30 specialists; staff decide

2 Scale-out

all regions, inside STEP planning; rollback = move @champion

First 90 days

- full ICR and PAD corpus behind the lessons tool
- amendment feed and a tender-package entity
- labelled evaluation set with specialists; shadow mode